

AI: SMPs QUIZ 2

One Shot Notes for 80+ Marks
by JSH SACH DEVA

Ques 1

29 November 2024 11:30

The distance matrix for 5 cities (A to E) is provided below. From each city the distances to other cities are listed in ascending order. For example, the distance from A to C is 56, from A to D is 80, and so on.

A	→ C: 56	D: 80	B: 110	E: 118
B	→ D: 40	E: 40	C: 80	A: 110
C	→ D: 40	A: 56	B: 80	E: 106
D	→ B: 40	C: 40	E: 72	A: 80
E	→ B: 40	D: 72	C: 106	A: 118

Solve the sub-questions using the TSP Branch-and-Bound algorithm.

Attention: Infer as much as possible (and as early as possible) about the permanent edges in the partial solutions. The semantics of the permanent edges (which are bidirectional) are as discussed in the practice assignments.

What is the lower bound on the cost of the tours (50) as per the TSP BnB algorithm discussed in class?

252

Take 2 min from each row & add all of them & then divide by 2 //

Infer all the permanently included edges in the node (S0, ~BD, ~BE) in the TSP BnB search tree. Enter the total number of permanently included edges in the text box.

3

1

- 1) To find the lower bound take 2 minimum from each row & add them all & divide the result by 2

$$\text{lower bound} = \frac{56 + 80 + 40 + 40 + 40 + 56 + 40 + 40 + 40 + 72}{2} = \frac{504}{2} = 252$$

- 2) To infer the permanently included edge we are given (S0, ~BD, ~BE) which means that B can't have edge with D & E

$$\therefore B-D \times \quad B-E \times$$

Hence only one possibility that B can have edge with A & C

$$\therefore A-B-C \rightarrow D-E$$

Hence no. of permanently included edges are

$$A-B, B-C, D-E = \underline{3} \text{ Ans}$$

- 3) Permanently excluded edges now;

So edges will be →

A	B	BC	CD	DE
AC		B	D	CE
AD		B	E	
AE				

Now we have to exclude the edge which can't be included from these

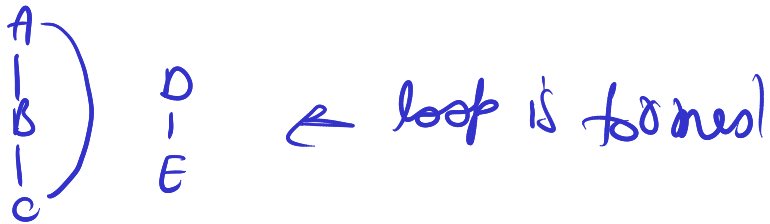
A
|
B
|
C

D
|
E

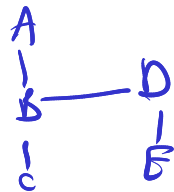
The criteria will be →

- 1) The edge shouldn't form a loop
- 2) The edge cannot make the in-degree / out-degree of a node > 1

for example AC cant be included
hence it will be an excluded edge



slly, BD will be an excluded edge



→ Here this wont be a TSP

~~AB~~ ~~BC~~ ~~CD~~ ~~DE~~
~~AC~~ ~~BD~~ ~~CE~~
~~AD~~ ~~BE~~
~~AE~~

Symbols → Included ✓

Excluded ✗

Hence the excluded edges are;
AC, BD, BE

therefore, there are 3 excluded edges in this question

PE → AC, BD, BE
3

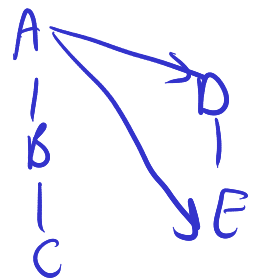
4) Now, to find the number of Tour in TSP

we can only have

A → D or A → E

to connect both the paths for complete tour;

Hence possible Number of Tours are 2



Infer all the permanently excluded edges in the node (SD-BD-BE) in the TSP BnB search tree. Enter the total number of permanently excluded edges in the text box.

3

How many tours are represented by the node (SD-BD-BE) in the TSP BnB search tree?

Now to find the cost of the Node (80, ~BD, ~BE)

Cost will be sum of 2 possible edges from all the nodes from this:



If there is only 1 edge connected in a node the 2nd edge will be chosen with the lowest cost

for eg:- A is connected to only B, A can be connected to D or E
 $\text{Cost}(AD) = 80$ $\text{Cost}(AE) = 118$

So AD will be chosen hence the 2 edges from A will be AB + AD

So from A → AB, AD

B → BA, BE (Not BD, BE because 2 are already there)

C → CB, CD (CE has higher cost than CD)

D → DE, DC (Not DB because it will not be TSB as indegree will be 2)

E → ED, EC (EA > EC in cost)

A → C: 56	D: 80	B: 110	E: 118
B → D: 40	E: 40	C: 80	A: 110
C → D: 40	A: 56	B: 80	E: 106
D → B: 40	C: 40	E: 72	A: 80
E → B: 40	D: 72	C: 106	A: 118

$$\text{So cost} = \frac{AB + AD + BA + BC + CB + CD + DE + DC + ED + EC}{2}$$

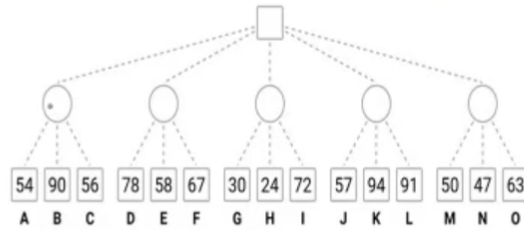
$$= \frac{110 + 80 + 110 + 80 + 80 + 40 + 72 + 40 + 72 + 106}{2}$$

$$= \frac{790}{2} = \underline{\underline{395}} \text{ Ans}$$

Ques - 2

List the horizon nodes in the best strategy.

Question Label : Comprehension



□ → max node

○ → min node

i> Solve the tree from bottom to top.

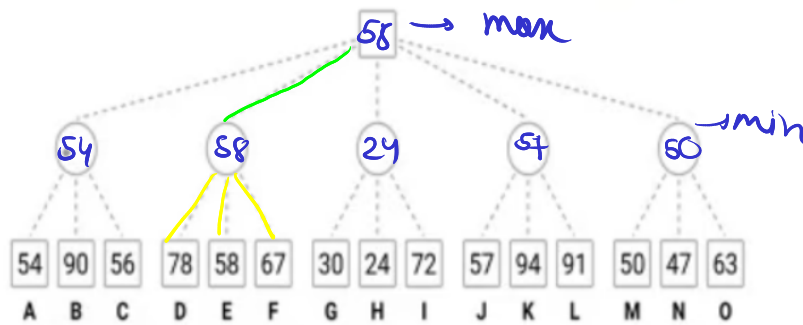
ii> For max node choose one branch

iii> For min node choose all branch.

List the horizon nodes in the best strategy.

D, E, F

○ → Give the minimum of all nodes ; □ → Give the maximum from all the nodes



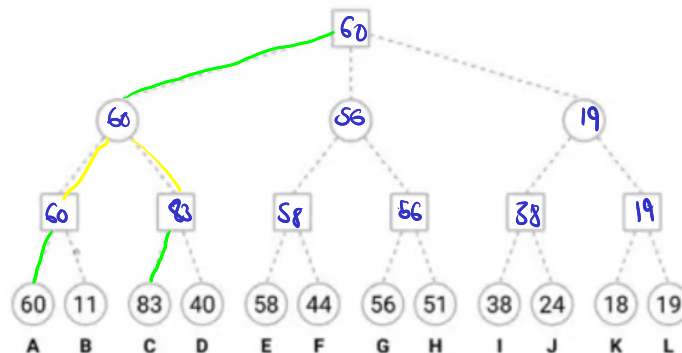
When at max node ; choose only one branch ;
When at min node ; choose all the branches ;

hence → First / then at min node choose all the branches

Hence it will stop at 3 nodes

D, E, F

Question Label : Comprehension



A, C

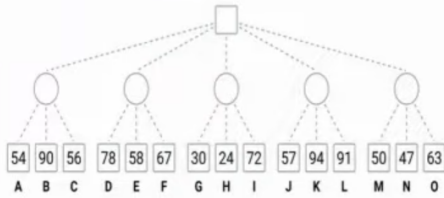
Ans

Ques-3

List the horizon nodes assigned SOLVED status by the SSS* algorithm.

22 November 2024 10:11

Question Label : Comprehension



A,D,E,F,G,J,M
A,D,G,J,M,E,F

i> we will make a priority queue first

ii> we will add initial nodes with cost in descending order.

iii> if we are at max node we will solve its adjacent sibling

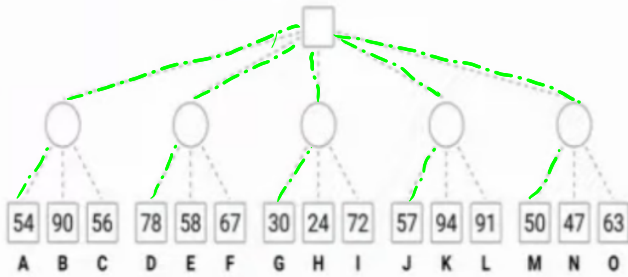
iv> if there are no more max node siblings we will mark it's parent min node solved.

v> if we are at min node we will directly pass the value to parent node. we will not solve min node adjacent sibling.

So, This a little complex question →

First to make the priority queue; mark all the lines from max node and then the left most line at min node

Question Label : Comprehension



Now for the priority queue; add all the max nodes with green. So it will be

A, D, G, J, M

Now solve the A Node using the priority & add the cost in descending order

D	78
J	57
A	54
M	50
G	30

Now Pop the first & start visiting the neighbours → again make the priority queue

So Pop D

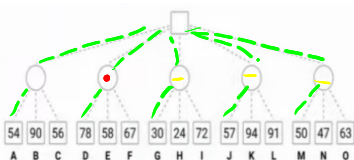
D	78
J	57
A	54
M	50
G	30

→ next neighbour to D is E with cost 58 so queue will be

E	58
J	57
A	54
M	50
G	30

Now Pop E & E's neighbour of E so here queue will be

Hence the graph will be



Now this is

→ solved
→ Skipped

Now Pop F

But no neighbour of F, so by rule

parent node will return to

F	67
J	57
A	54
M	50
G	30

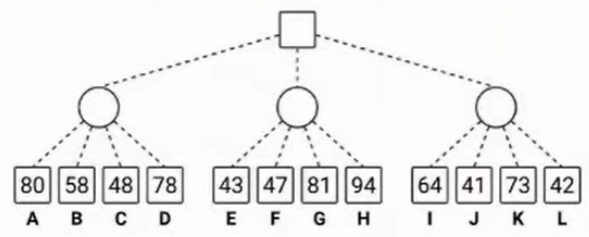
So we will skip the adjacent min node if one min node is solved & hence the nodes are

A, D, G, J, M, E, F Now for the answer you can arrange this in ascending order

A, D, E, F, G, J, M

~~the~~ * Just follow the 5 rules in the question written in red

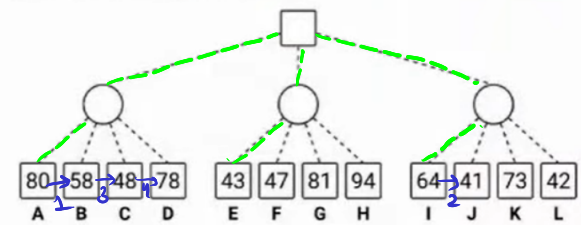
Question Label : Comprehension



A,B,C,D,E,I,J
A,E,I,B,J,C,D

Now make the priority queue;

Question Label : Comprehension



A	80
I	64
E	43

①

→

I	64
B	58
E	43

②

→

B	58
F	47
J	41

③

→

C	48
E	43
J	41

④

→

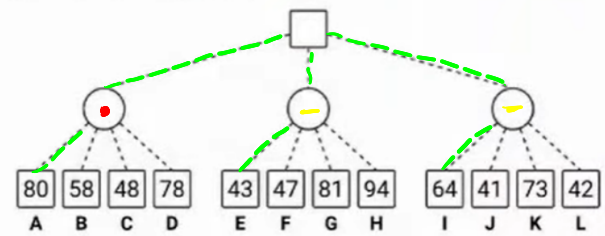
D	78
E	43
J	41

⑤

priority queue will be A, E, I

✗ No neighbors & D 80 max Node D is solved

Question Label : Comprehension



hence the graph after nodes will be ;

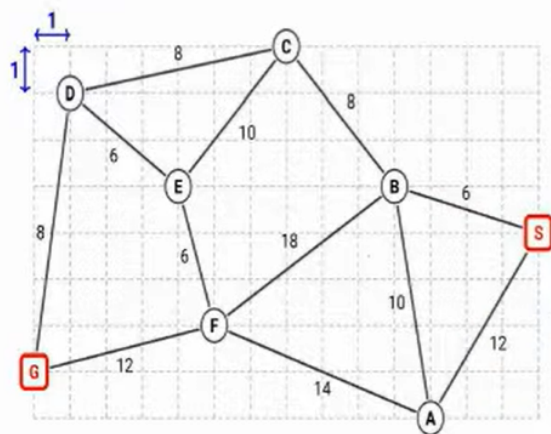
A, E, I, B, J, G, D
Now arrange the answer

A, B, C, D, E, I, J

Ques-4

Emulate A*, wA* and Branch-and-Bound on the given map, then answer the given subquestions

25 November 2024 15:27



heuristic values
 S: 16 A: 12 D: 7
 G: 0 B: 14 E: 8
 C: 14 F: 6

$$F(x) = g(x) + h(x)$$

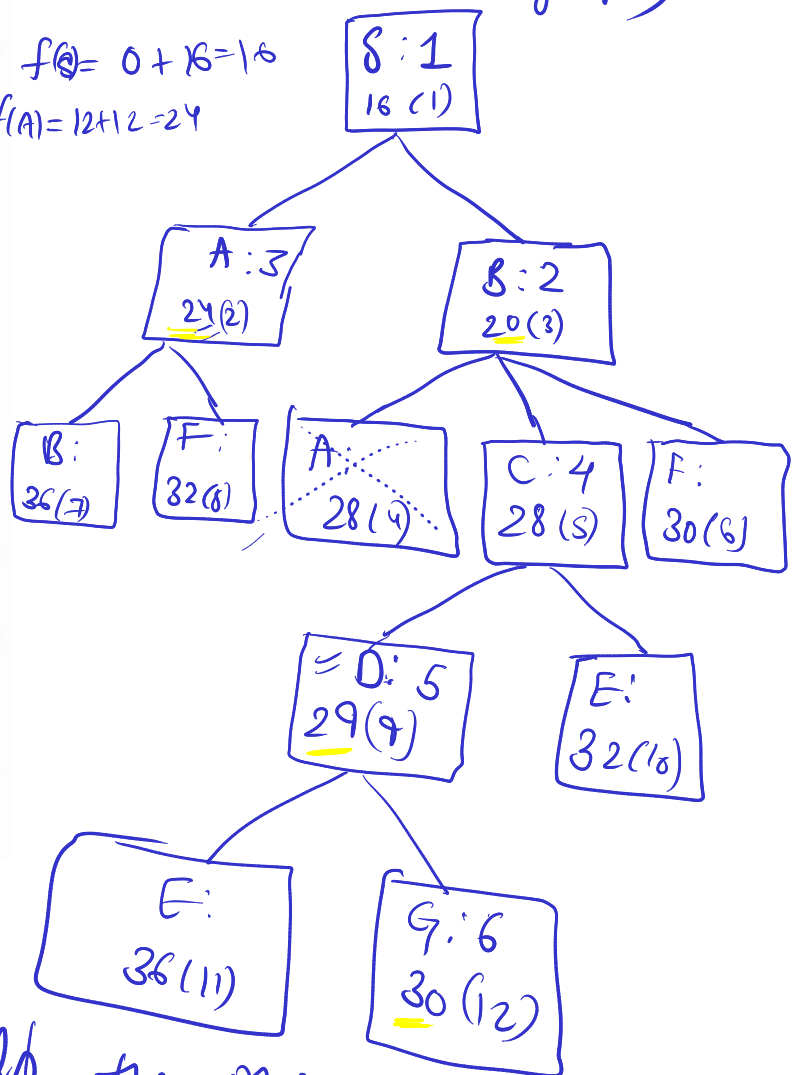
In the map, S is the first node to be inspected, determine the next 4 nodes (from the 2nd to 5th node) inspected by A*.

B, A, C, D

Now make the graph

$$f(S) = 0 + 16 = 16$$

$$f(A) = 12 + 12 = 24$$



Steps: Next step would be the one with minimum $f(x)$
 $f(x) = g(x) + h(x)$; $g(x)$ is the cost to get to the node from the source node S
 for eg

$$g(D) = SB + BC + CD = 6 + 8 + 8 = 22$$

$$f(D) = 22 + 7 = 29$$



heuristic value

We have rejected

A:
28(4)

because we have the

step A:
24(2)

already & it also had the lowest $f(x)$
 here we find the path from

that

Now the first 4 nodes will be from the step 2 to 5 is B, A, C, D

In the map, S is the first node to be inspected, determine the next 4 nodes (from the 2nd to 5th node) inspected by A*. Enter the nodes in the order they were inspected.

B, A, C, D

For the 4 nodes from the 2nd to 5th node, list the f-values of those nodes as a comma separated list.

20, 24, 28, 29

For the 4 nodes from the 2nd to 5th node, list the parent nodes assigned by A*.

S, S, B, C

What is the final path found by A*?

S, B, C, D, G

For w=2, what is the final path found by wA* algorithm?

S, B, F, G

Now for 2nd question we need to give one answer for

$f(B), f(A), f(C), f(D)$

20, 24, 28, 29

Now parent of $\text{parent}(B, A, C, D)$

= S, S, B, C

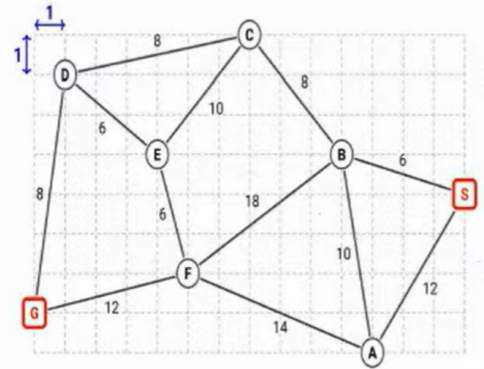
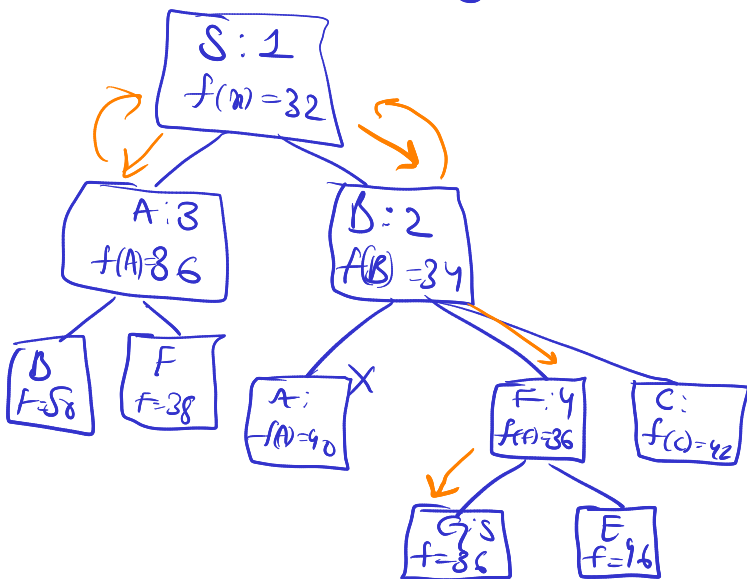
Now final path found by it
S, B, C, D, G

for $W A^*$

$$f(x) = g(x) + w h(x)$$

hence

$$f(x) = g(x) + 2h(x)$$



heuristic values			
S: 16	A: 12	D: 7	
G: 0	B: 14	E: 8	
	C: 14	F: 6	

$$f(x) = g(x) + h(x)$$

$2 * h(x)$			
S: 32	A: 24	D: 14	
G: 0	B: 28	E: 16	
	C: 28	F: 12	

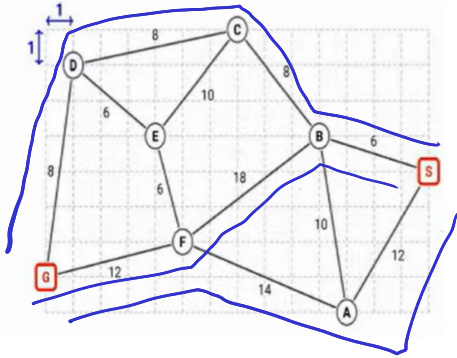
So path found by $W A^*$ is S, B, F, G

What is the cost of the path found by Branch-and-Bound algorithm? Use the Branch-and-Bound version that PREVENTS cyclic expansions like S,A,S,A,S,A,...

30

Trick for Branch & Bound

Path is Just add the shortest path by the cost that will be the answer



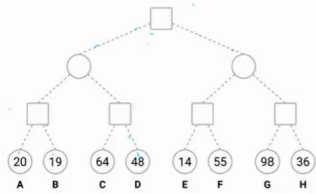
So S B C D G has cost 30
(6+8+8+8)

S A F G has cost 38 (12+14+12)

S B F G has cost 36 (6+18+12)

hence lowest cost is 30

List the horizon nodes in the initial cluster formed by SSS* algorithm.

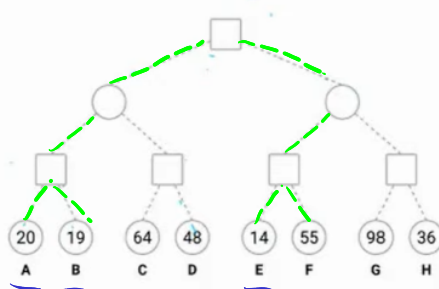


A, B, E, F

Ques 5

Rules $\square \rightarrow$ max node $\circ \rightarrow$ min node
For max node $\rightarrow$ select all branch
For min node $\rightarrow$ select only one that is left most one,

List the horizon nodes in the initial cluster formed by SSS* algorithm.

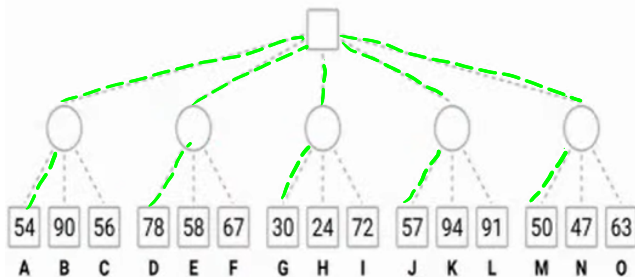


So the end nodes will be $\rightarrow$ A, B, E, F

$\square \rightarrow$ max node
 $\circ \rightarrow$ min node

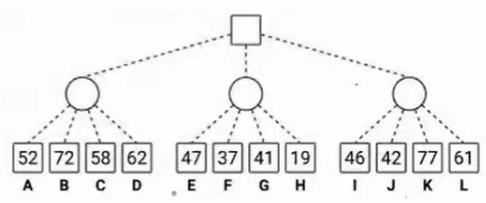
So the nodes will be $\rightarrow$

A, D, G, J, M



Ques-6

GAMES - AlphaBeta, SSS*



if N is a max node
 if $\alpha \geq \beta$ return β
 called β -cutoff
 else
 return α
 if N is a min node
 if $\alpha \geq \beta$ return α
 called α -cutoff
 else
 return β

List the horizon nodes pruned by Alpha-Beta algorithm.

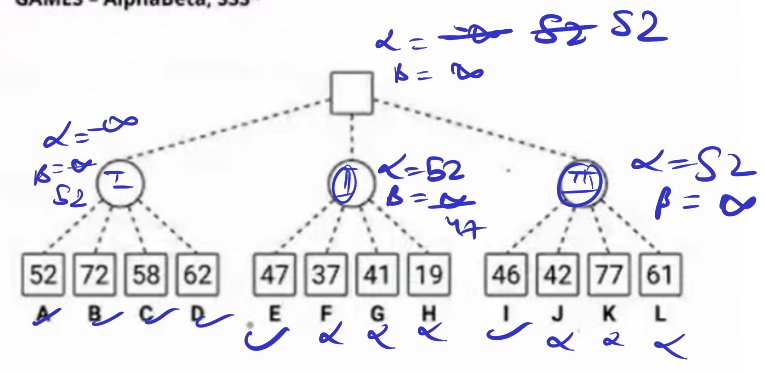
F, G, H, J, K, L

Here in this question set $\alpha = -\infty$ & $\beta = \infty$
 $\square \rightarrow$ update α
 $\bigcirc \rightarrow$ update β

So go to first min node

So in $\textcircled{I}$, it is a min node and at min node we will update β here β will take A and has value 52 then So β will be updated to 52

GAMES - AlphaBeta, SSS*



then $72 < 52 \times 58 < 52 \times 62 < 52 \times$ hence $\textcircled{I}$ will return $\beta = 52$ to the max node & at $\square$ or max node α is updated
 $\therefore \boxed{\alpha = 52 \text{ \& } \beta = \infty}$

Now at $\textcircled{II}$ $\beta = \infty$ & updated to $\beta = E = 47$

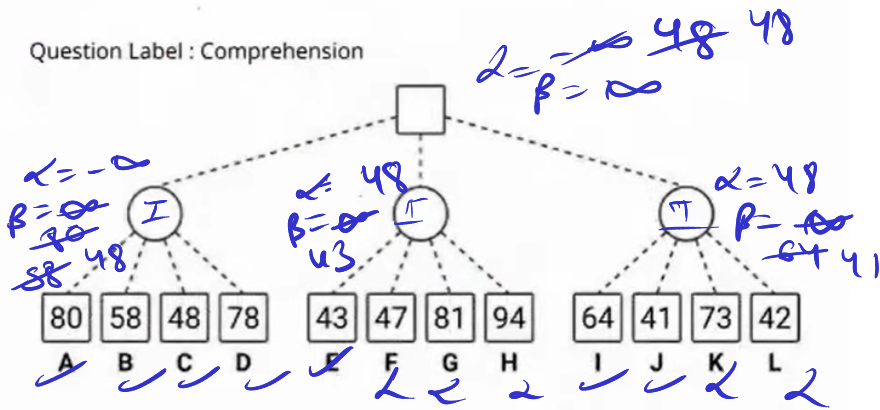
Now $\alpha \geq \beta$ or $52 \geq 47$ Hence it will return α to the max node & α would again get value 52 & $\beta = \infty$

again at $\textcircled{III}$ $\alpha = 52, \beta = \infty$; now β will be updated to $\beta = I = 46 \therefore \alpha = 52, \beta = 46 \Rightarrow \alpha \geq \beta \therefore$

min node will return $\boxed{\alpha = 52}$ & it will finish

Now, the unvisited nodes are F, G, H, J, K, L
 $\rightarrow$ answer will be F, G, H, J, K, L

Question Label : Comprehension



So in (I) β from ∞
to $80 \rightarrow 58 \rightarrow \boxed{48}$

Now $\alpha \geq \beta$

$-\infty \geq 48 \times$

return $\beta = 48$

at max node α is updated to $\alpha = 48$
 $\beta = \infty$

now at (II) $\alpha = 48, \beta = \infty$
 $\beta \rightarrow 43$

Now $\alpha \geq \beta$ or $48 \geq 43$ hence return α

$\therefore \alpha$ is updated to again 48

now at (III) $\alpha = 48, \beta = \infty$

$\beta \rightarrow 64$; $48 \geq 64$ hence β will
be updated
 $\beta \rightarrow 64 \rightarrow 41$ again

$48 \geq 41 \therefore \alpha \geq \beta$ at min node

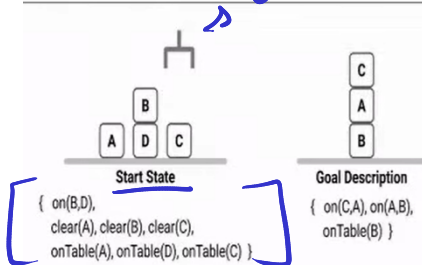
hence α is updated
to $\alpha = 41$ & $\beta = \infty$

So Non visited nodes are $\rightarrow$ F, G, H, K, L ans

arm empty

Ques 7

Precondition(Action) $\subseteq$ Start State



PREDICATES

armEmpty The arm is not holding any block, it is empty.
 holding(X) The arm is holding X.
 onTable(X) X is on the table.
 clear(X) X has nothing above it, it is clear.
 on(X,Y) X is directly on Y.

OPERATORS

Pickup(X): pick up X from the table
 Preconditions: { armEmpty, clear(X), onTable(X) }
 Add Effects : { holding(X) }
 Del Effects : { armEmpty, onTable(X) }

Putdown(X): place X on the table.
 Preconditions: { holding(X) }
 Add Effects : { armEmpty, onTable(X) }
 Del Effects : { holding(X) }

Unstack(X,Y): pick up X that is directly sitting on Y.
 Preconditions: { armEmpty, clear(X), on(X,Y) }
 Add Effects : { clear(Y), holding(X) }
 Del Effects : { armEmpty, on(X,Y) }

Stack(X,Y): place X directly on top of Y.
 Preconditions: { holding(X), clear(Y) }
 Add Effects : { armEmpty, on(X,Y) }
 Del Effects : { holding(X), clear(Y) }

Which of the following are applicable actions in the start state for the given planning problem?

Options:

- 6406532878284. ✓ Pickup(A)
- 6406532878285. ✓ Pickup(C)
- 6406532878286. ✗ Stack(A,B)
- 6406532878287. ✗ Stack(C,A)
- 6406532878288. ✓ Unstack(B,D)

6406532878289. ✗ Unstack(A,B)

In this we need to check every option + Put those values in the function

Pickup(A) → Preconditions { armEmpty, clear(A), onTable(A) }

So yes from starting conditions arm empty ✓
 clear(A) ✓
 onTable(A) ✓

Sly;

Pickup(C) → Preconditions { armEmpty, clear(C), onTable(C) }

So yes from starting conditions arm empty ✓
 clear(C) ✓
 onTable(C) ✓

Sly; Stack(A,B) → Preconditions (holding(A), clear(B))
 holding(A) ✓
 clear(B) ✗

Hence Preconditions are not satisfied ✗

Stack(C, A) $\rightarrow$ Precondition (holding(C), clear(A))

holding(C) ✓

clear(A) ✗

Hence Precondition are not satisfied ✗

Unstack(B, D) = precondition (armempty, clear(B),

armempty ✓

on(A, B))

clear(B) ✓

on(B, D) ✓

Hence satisfied ✓

Unstack(A, B) = precondition (armempty, clear(A),

armempty ✓

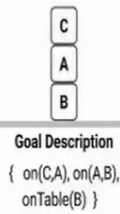
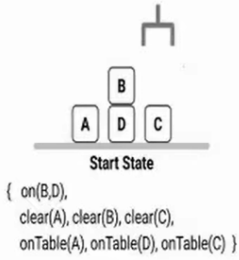
on(A, B))

clear(✓) ✓

on(A, B) ✗

Hence Not satisfied ✗

Ques-8



PREDICATES

armEmpty The arm is not holding any block, it is empty.
holding(X) The arm is holding X.
onTable(X) X is on the table.
clear(X) X has nothing above it, it is clear.
on(X,Y) X is directly on Y.

OPERATORS

Pickup(X): pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }
Add Effects : { holding(X) }
Del Effects : { armEmpty, onTable(X) }

Putdown(X): place X on the table.

Preconditions: { holding(X) }
Add Effects : { armEmpty, onTable(X) }
Del Effects : { holding(X) }

Unstack(X,Y): pick up X that is directly sitting on Y.

Preconditions: { armEmpty, clear(X), on(X,Y) }
Add Effects : { clear(Y), holding(X) }
Del Effects : { armEmpty, on(X,Y) }

Stack(X,Y): place X directly on top of Y.

Preconditions: { holding(X), clear(Y) }
Add Effects : { armEmpty, on(X,Y) }
Del Effects : { holding(X), clear(Y) }

For relevant actions

$$\rightarrow \text{Goal Description} \cap \text{effect}^+(Action) \neq \emptyset$$

$$\text{Goal Description} \cap \text{effect}^-(Action) = \emptyset$$

Which of the following are **relevant** actions for the goal description in the given planning problem?

Options:

6406532878290. ✖ Pickup(B)
6406532878291. ✖ Pickup(C)
6406532878292. ✔ Putdown(B)
6406532878293. ✔ Stack(A,B)
6406532878294. ✔ Stack(C,A)
6406532878295. ✖ Unstack(C,A)

Now in this one we need to check all the options & an option will be correct if

$$\text{goal description} \cap \text{add effects} \neq \emptyset$$

So, both should be true if an option is correct

$$\text{goal description} \cap \text{del effects} = \emptyset$$

So Pickup(B) $\rightarrow \text{GD} \cap \text{add effects} = \emptyset$
 $\rightarrow \text{GD} \cap \{ \text{holding(B)} \} = \emptyset$
 False

Pickup(C) $\rightarrow \text{GD} \cap \{ \text{holding(C)} \} = \emptyset$ ✖

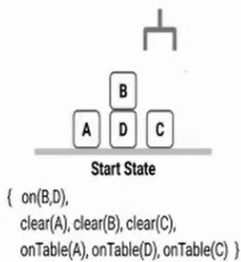
Putdown() $\rightarrow \text{GD} \cap \{ \text{arm empty, onTable(B)} \} = \{ \text{onTable(B)} \} \neq \emptyset$
 True

$\text{GD} \cap \{ \text{holding(B)} \} = \emptyset \rightarrow \text{True}$
 $T \vee T = T$

So check for all the options correct option

Ques - 9

What is the length of the optimal plan? Enter the number of actions in the optimal plan.



PREDICATES

armEmpty The arm is not holding any block, it is empty.
 holding(X) The arm is holding X.
 onTable(X) X is on the table.
 clear(X) X has nothing above it, it is clear.
 on(X,Y) X is directly on Y.

OPERATORS

Pickup(X): pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }

Add Effects : { holding(X) }

Del Effects : { armEmpty, onTable(X) }

Goal State mai ba Stacke hona chahiye

B - Pickup
 B - Putdown
 A - Pickup
 A - Putdown
 C - Pick
 C - Putdown

6 A

very simple question, answer will be the no. of steps to reach from start state to Goal state

So first 1) Pickup (B)

2) Put down (B)

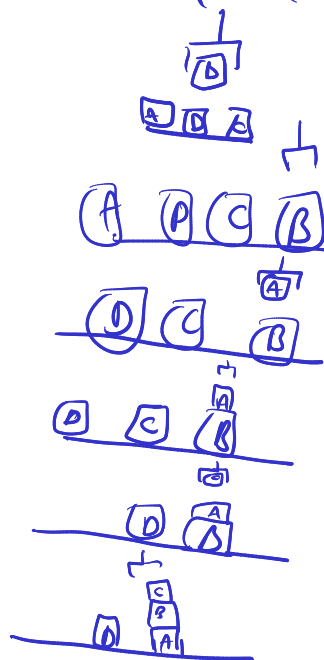
3) Pickup (A)

4) Putdown (A)

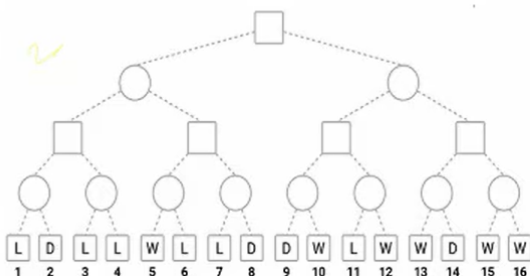
5) Pickup (C)

6) Putdown (C)

So ans is 6



The figure shows a game tree with evaluations W (win), D (draw) and L (loss) from Max's perspective. The horizon nodes carry evals (W/D/L) and node labels (1 to 16).

Ques - 10

max

W → D → L

min node

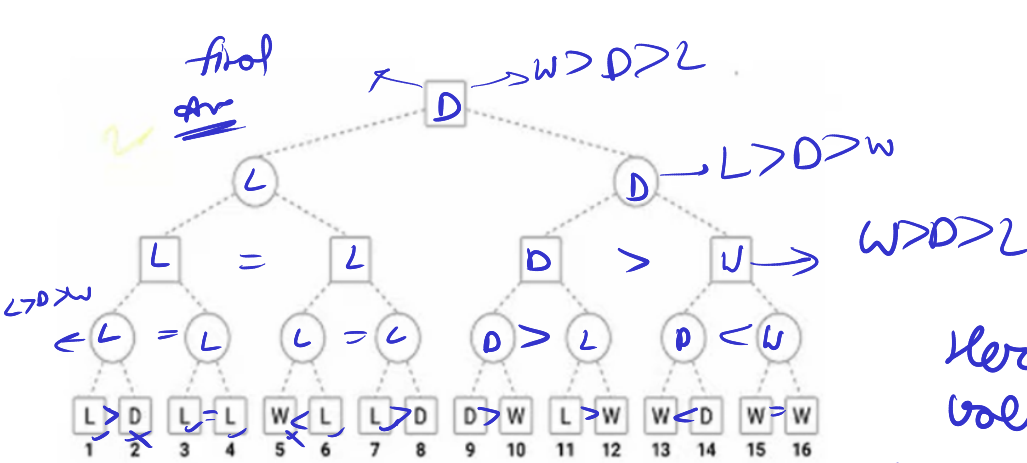
L → D → W

Priorities

O → min

□ → max

What is the outcome (W, D or L) of the game when both players play perfectly?



W - win
D - Draw
L - Loss

Here we need to find value of parent node & the value of parent node will be chosen from the so for min node $L > D > W$ for max hence we select

node priority is like that..

Node priority is $W > D > L$

Change the value of only one horizon node such that the outcome of the game changes. Which horizon node will you change and what will be its new value? Enter the label of the horizon node and the new value as a comma separated list.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.

Answer format: 16,X

Response Type : Alphanumeric
Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

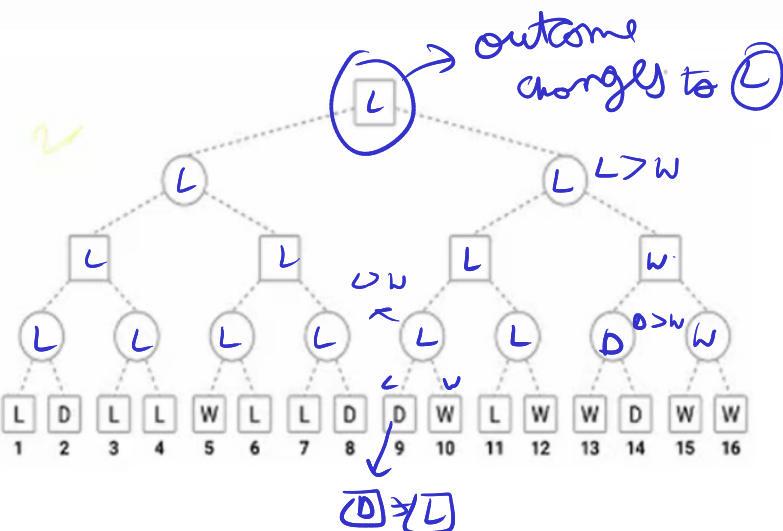
Possible Answers :

9,W

9,L

10,L

11,W



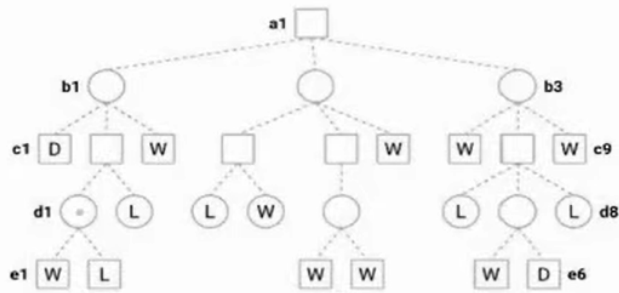
So very simple; it's asking if you could change value of one node & whole outcome change like from D to W or L what node would that be & what outcome

So if I change 9D to 9L

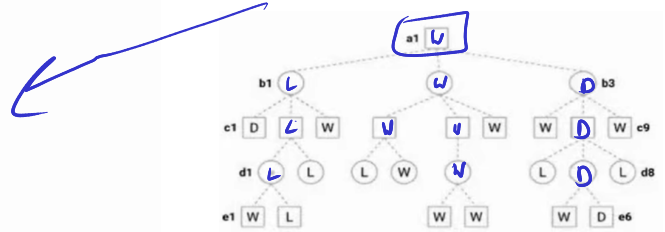
hence ans is 9,L

& possible 3 more answers are there

The figure shows a game tree with evaluations W (win), L (loss) and D (draw) from Max's perspective. The nodes are labeled in a level-by-level (a,b,...,e) left-to-right (1,2,...) manner as indicated in the game tree.



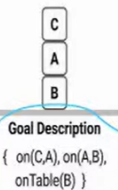
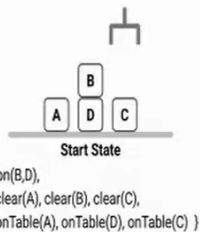
Now solve this



What is the outcome (W, D or L) of the game when both players play perfectly?

for W

Que - Lost



For the subgoal ordering given in the goal description (and using the given tie breaking rules) which of the following can be pushed as the first four elements onto the stack by the Goal Stack Planning algorithm? In the representation below, the bottom of the stack is on the right end, marked by BOTTOM.

- Options :
- 6406532878296. * { on(C,A), on(A,B), onTable(B) }; on(C,A); on(A,B); onTable(B); BOTTOM
 - 6406532878297. * { on(C,A), on(A,B), onTable(B) }; onTable(B); on(A,B); on(C,A); BOTTOM
 - 6406532878298. * on(C,A); on(A,B); onTable(B); { on(C,A), on(A,B), onTable(B) }; BOTTOM
 - 6406532878299. * onTable(B); on(A,B); on(C,A); { on(C,A), on(A,B), onTable(B) }; BOTTOM

PREDICATES

armEmpty The arm is not holding any block, it is empty.
 holding(X) The arm is holding X.
 onTable(X) X is on the table.
 clear(X) X has nothing above it, it is clear.
 on(X,Y) X is directly on Y.

OPERATORS

Pickup(X): pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }
 Add Effects : { holding(X) }
 Del Effects : { armEmpty, onTable(X) }

Here in this question ; for BOTTOM is that answer will be in the format

for
 So
 goal Description = { on(C,A), on(A,B), onTable(B) }
 for
 onTable(B); on(A,B); on(C,A); { on(C,A), on(A,B), onTable(B) }; BOTTOM